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Sets and Relations

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 92 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Consider the relation R on the set $\{-2,-1,0,1,2\}$ defined by $(a,b) \in R$ if and only if $1+ab>0$. Then, among the statements :

I. The number of elements in R is 17

II. R is an equivalence relation

(A) Only I is true
(B) Only II is true
(C) Both I and II are true
(D) Neither I nor II is true

Open solution

Q2. JEE Main 2026 (Online) 24th January Morning Shift

Let R be a relation defined on the set $\{1,2,3,4\} \times \{1,2,3,4\}$ by

$\mathrm{R}=\{((a,b),(c,d)) : 2a+3b=3c+4d\}$.

Then the number of elements in R is

(A) 6 (B) 15 (C) 12 (D) 18

Open solution

Q3. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\mathrm{A}=\{0,1,2,\ldots,9\}$. Let R be a relation on A defined by $(x,y) \in \mathrm{R}$ if and only if $|x-y|$ is a multiple of 3.

Given below are two statements :

Statement I : $n(\mathrm{R})=36$.

Statement II : R is an equivalence relation.

In the light of the above statements, choose the ***correct*** answer from the options given below :

(A) Statement I is correct but Statement II is incorrect
(B) Both Statement I and Statement II are correct
(C) Both Statement I and Statement II are incorrect
(D) Statement I is incorrect but Statement II is correct

Open solution

Q4. JEE Main 2026 (Online) 23rd January Morning Shift

Let $\mathrm{A}=\{-2,-1,0,1,2,3,4\}$. Let R be a relation on A defined by $x\mathrm{R}y$ if and only if $2x+y \leqslant 2$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then $l+\mathrm{m}+\mathrm{n}$ is equal to :

(A) 34 (B) 32 (C) 33 (D) 35

Open solution

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Q5. JEE Main 2026 (Online) 22nd January Evening Shift

The number of elements in the relation $\mathrm{R}=\left\{(x, y): 4x^2+y^2<52, x, y \in \mathbf{Z}\right\}$ is

(A) 86 (B) 67 (C) 89 (D) 77

Open solution

Q6. JEE Main 2026 (Online) 22nd January Morning Shift

Let the relation R on the set $\mathrm{M}=\{1,2,3, \ldots, 16\}$ be given by $\mathrm{R}=\{(x, y): 4y=5x-3, x, y \in \mathrm{M}\}$.

Then the minimum number of elements required to be added in R , in order to make the relation symmetric, is equal to

(A) 4 (B) 3 (C) 1 (D) 2

Open solution

Q7. JEE Main 2026 (Online) 21st January Evening Shift

Let $A=\{x:|x^2-10| \leq 6\}$ and $B=\{x:|x-2|>1\}$. Then

(A) $A \cup B=(-\infty, 1] \cup(2, \infty)$ (B) $B-A=(-\infty,-4) \cup(-2,1) \cup(4, \infty)$

(C) $A-B=[2,3)$ (D) $A \cap B=[-4,-2] \cup[3,4]$

Open solution

Q8. JEE Main 2026 (Online) 21st January Evening Shift

Let $A=\{2,3,5,7,9\}$. Let R be the relation on A defined by xRy if and only if $2x \leq 3y$. Let l be the number of elements in R, and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then $l+m$ is equal to:

(A) 21 (B) 25 (C) 23 (D) 27

Open solution

Q9. JEE Main 2026 (Online) 21st January Morning Shift

The number of relations, defined on the set $\{\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}\}$, which are both reflexive and symmetric, is equal to:

(A) 16 (B) 64 (C) 256 (D) 1024

Open solution

Q10. JEE Main 2025 (Online) 8th April Evening Shift

Let **A** = {0, 1, 2, 3, 4, 5}. Let **R** be a relation on **A** defined by (x, y) ∈ **R** if and only if max{x, y} ∈ {3, 4}. Then among the statements

(S_1): The number of elements in **R** is 18, and

(S_2): The relation **R** is symmetric but neither reflexive nor transitive

(A) both are false (B) only (S_1) is true

(C) only (S_2) is true (D) both are true

Open solution

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Q11. JEE Main 2025 (Online) 7th April Evening Shift

Let **A** = $\{ (\alpha, \beta) \in \mathbb{R} \times \mathbb{R} : |\alpha - 1| \leq 4 \text{ and } |\beta - 5| \leq 6 \}$

and **B** = $\{ (\alpha, \beta) \in \mathbb{R} \times \mathbb{R} : 16(\alpha - 2)^2 + 9(\beta - 6)^2 \leq 144 \}$.

Then

(A) A $\subset$ B

(B) B $\subset$ A

(C) neither A $\subset$ B nor B $\subset$ A

(D) $A \cup B = \{(x, y) : -4 \leqslant x \leqslant 4, -1 \leqslant y \leqslant 11\}$

Open solution

Q12. JEE Main 2025 (Online) 4th April Evening Shift

Let $\mathrm{A} = \{-3, -2, -1, 0, 1, 2, 3\}$ and R be a relation on A defined by $x\mathrm{R}y$ if and only if $2x - y \in \{0, 1\}$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l + \mathrm{m} + \mathrm{n}$ is equal to:

(A) 17

(B) 18

(C) 15

(D) 16

Open solution

Q13. JEE Main 2025 (Online) 4th April Morning Shift

Consider the sets $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 = 25\}$, $B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + 9y^2 = 144\}$, $C = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 4\}$ and $D = A \cap B$. The total number of one-one functions from the set D to the set C is:

(A) 15120

(B) 18290

(C) 17160

(D) 19320

Open solution

Q14. JEE Main 2025 (Online) 3rd April Evening Shift

Let $A = \{-2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by $x\mathrm{R}y$ if and only if $y = \max\{x, 1\}$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l + m + n$ is equal to

(A) 11

(B) 12

(C) 14

(D) 13

Open solution

Q15. JEE Main 2025 (Online) 3rd April Morning Shift

Let $\mathrm{A} = \{-3, -2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by $x\mathrm{R}y$ if and only if $0 \leq x^2 + 2y \leq 4$. Let l be the number of elements in R and m be the minimum number of elements required to be added in R to make it a reflexive relation. Then $l + m$ is equal to

(A) 18

(B) 20

(C) 17

(D) 19

Open solution

Q16. JEE Main 2025 (Online) 2nd April Evening Shift

Let $A = \{1, 2, 3, \ldots, 100\}$ and R be a relation on A such that $R = \{(a, b) : a = 2b + 1\}$. Let $(a_1, a_2), (a_2, a_3), (a_3, a_4), \ldots, (a_k, a_{k+1})$ be a sequence of k elements of R such that the second entry of an ordered pair is equal to the first entry of the next ordered pair. Then the largest integer k , for which such a sequence exists, is equal to :

(A) 6

(B) 8

(C) 7

(D) 5

Open solution

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Q17. JEE Main 2025 (Online) 2nd April Morning Shift

Let A be the set of all functions $f : \mathbf{Z} \to \mathbf{Z}$ and R be a relation on A such that $\mathrm{R} = \{(\mathrm{f}, \mathrm{g}) : f(0) = \mathrm{g}(1) \text{ and } f(1) = \mathrm{g}(0)\}$. Then R is :

(A) Symmetric and transitive but not reflexive

(B) Symmetric but neither reflexive nor transitive

(C) Transitive but neither reflexive nor symmetric

(D) Reflexive but neither symmetric nor transitive

Open solution

Q18. JEE Main 2025 (Online) 29th January Evening Shift

Let $\mathrm{S} = \mathbf{N} \cup \{0\}$. Define a relation R from S to $\mathbf{R}$ by :

$\mathrm{R} = \left\{(x, y) : \log_{\mathrm{e}} y = x \log_{\mathrm{e}} \left(\frac{2}{5}\right), x \in \mathrm{S}, y \in \mathbf{R}\right\}$.

Then, the sum of all the elements in the range of R is equal to :

(A) $\frac{3}{2}$ (B) $\frac{10}{9}$ (C) $\frac{5}{2}$ (D) $\frac{5}{3}$

Open solution

Q19. JEE Main 2025 (Online) 29th January Morning Shift

Define a relation R on the interval $\left[0, \frac{\pi}{2}\right)$ by $x \,\mathrm{R}\, y$ if and only if $\sec^2 x - \tan^2 y = 1$. Then R is :

(A) both reflexive and symmetric but not transitive

(B) both reflexive and transitive but not symmetric

(C) reflexive but neither symmetric nor transitive

(D) an equivalence relation

Open solution

Q20. JEE Main 2025 (Online) 28th January Morning Shift

The relation $R = \{(x, y) : x, y \in \mathbb{Z} \text{ and } x + y \text{ is even }\}$ is:

(A) reflexive and transitive but not symmetric

(B) reflexive and symmetric but not transitive

(C) an equivalence relation

(D) symmetric and transitive but not reflexive

Open solution

Q21. JEE Main 2025 (Online) 24th January Evening Shift

Let $\mathrm{A} = \left\{x \in (0, \pi) - \left\{\frac{\pi}{2}\right\} : \log_{(2/\pi)} |\sin x| + \log_{(2/\pi)} |\cos x| = 2\right\}$ and $\mathrm{B} = \{x \geqslant 0 : \sqrt{x}(\sqrt{x} - 4) - 3|\sqrt{x} - 2| + 6 = 0\}$. Then $\mathrm{n}(\mathrm{A} \cup \mathrm{B})$ is equal to :

(A) 4 (B) 8 (C) 6 (D) 2

Open solution

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Q22. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Let $X = \mathbf{R} \times \mathbf{R}$. Define a relation R on X as :

$(a_1, b_1) R (a_2, b_2) \Leftrightarrow b_1 = b_2$

Statement I: R is an equivalence relation.

Statement II : For some $(a, b) \in X$, the set $S = \{(x, y) \in X : (x, y) R (a, b)\}$ represents a line parallel to $y = x$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

Q23. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Let $A = \{(x, y) \in \mathbf{R} \times \mathbf{R} : |x + y| \geqslant 3\}$ and $B = \{(x, y) \in \mathbf{R} \times \mathbf{R} : |x| + |y| \leq 3\}$. If $C = \{(x, y) \in A \cap B : x = 0 \text{ or } y = 0\}$, then $\sum_{(x,y) \in C} |x + y|$ is :

(A) 18 (B) 24 (C) 15 (D) 12

Q24. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Let $R = \{(1, 2), (2, 3), (3, 3)\}$ be a relation defined on the set $\{1, 2, 3, 4\}$. Then the minimum number of elements, needed to be added in R so that R becomes an equivalence relation, is:

(A) 9 (B) 8 (C) 7 (D) 10

Q25. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Let $A = \{1, 2, 3, \ldots, 10\}$ and $B = \left\{\frac{m}{n} : m, n \in A, m < n \text{ and } \gcd(m, n) = 1\right\}$. Then $n(B)$ is equal to :

(A) 29 (B) 31 (C) 37 (D) 36

Q26. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

The number of non-empty equivalence relations on the set $\{1, 2, 3\}$ is :

(A) 7 (B) 4 (C) 5 (D) 6

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Q27. JEE Main 2024 (Online) 8th April Evening Shift

Let $A = \{2, 3, 6, 8, 9, 11\}$ and $B = \{1, 4, 5, 10, 15\}$. Let R be a relation on $A \times B$ defined by $(a, b)R(c, d)$ if and only if $3ad - 7bc$ is an even integer. Then the relation R is

(A) reflexive but not symmetric.

(B) an equivalence relation.

(C) reflexive and symmetric but not transitive.

(D) transitive but not symmetric.

Open solution

Q28. JEE Main 2024 (Online) 6th April Evening Shift

Let $\mathrm{A} = \{1, 2, 3, 4, 5\}$. Let R be a relation on A defined by $x\mathrm{R}y$ if and only if $4x \leq 5\mathrm{y}$. Let m be the number of elements in R and n be the minimum number of elements from $\mathrm{A} \times \mathrm{A}$ that are required to be added to R to make it a symmetric relation. Then m + n is equal to :

(A) 23 (B) 26 (C) 25 (D) 24

Open solution

Q29. JEE Main 2024 (Online) 6th April Morning Shift

Let $A = \{n \in [100, 700] \cap \mathrm{N} : n$ is neither a multiple of 3 nor a multiple of $4\}$. Then the number of elements in A is

(A) 300 (B) 310 (C) 290 (D) 280

Open solution

Q30. JEE Main 2024 (Online) 6th April Morning Shift

Let the relations R_1 and R_2 on the set $X = \{1, 2, 3, \ldots, 20\}$ be given by $R_1 = \{(x, y) : 2x - 3y = 2\}$ and $R_2 = \{(x, y) : -5x + 4y = 0\}$. If M and N be the minimum number of elements required to be added in R_1 and R_2, respectively, in order to make the relations symmetric, then $M + N$ equals

(A) 16 (B) 12 (C) 8 (D) 10

Open solution

Q31. JEE Main 2024 (Online) 4th April Evening Shift

Let a relation R on $\mathbb{N} \times \mathbb{N}$ be defined as: $(x_1, y_1)\mathrm{R}\,(x_2, y_2)$ if and only if $x_1 \leq x_2$ or $y_1 \leq y_2$. Consider the two statements:

(I) R is reflexive but not symmetric.

(II) R is transitive

Then which one of the following is true?

(A) Only (II) is correct. (B) Both (I) and (II) are correct.

(C) Neither (I) nor (II) is correct. (D) Only (I) is correct.

Open solution

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Q32. JEE Main 2024 (Online) 1st February Evening Shift

Consider the relations R_1 and R_2 defined as $aR_1b \Leftrightarrow a^2 + b^2 = 1$ for all $a, b \in \mathbf{R}$ and $(a, b)R_2(c, d) \Leftrightarrow a + d = b + c$ for all $(a, b), (c, d) \in \mathbf{N} \times \mathbf{N}$. Then :

(A) R_1 and R_2 both are equivalence relations

(B) Only R_1 is an equivalence relation

(C) Only R_2 is an equivalence relation

(D) Neither R_1 nor R_2 is an equivalence relation

Open solution

Q33. JEE Main 2024 (Online) 29th January Evening Shift

If R is the smallest equivalence relation on the set $\{1, 2, 3, 4\}$ such that $\{(1, 2), (1, 3)\} \subset \mathrm{R}$, then the number of elements in R is _____.

(A) 15 (B) 10 (C) 12 (D) 8

Open solution

Q34. JEE Main 2024 (Online) 29th January Morning Shift

Let R be a relation on $Z \times Z$ defined by $(a, b)R(c, d)$ if and only if $ad - bc$ is divisible by 5. Then R is

(A) Reflexive and transitive but not symmetric

(B) Reflexive and symmetric but not transitive

(C) Reflexive but neither symmetric nor transitive

(D) Reflexive, symmetric and transitive

Open solution

Q35. JEE Main 2024 (Online) 27th January Evening Shift

Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point $P(m, n)$ from the point $Q(-2, -3)$ is :

(A) 8 (B) 10 (C) 4 (D) 6

Open solution

Q36. JEE Main 2024 (Online) 27th January Morning Shift

Let $S = \{1, 2, 3, \ldots, 10\}$. Suppose M is the set of all the subsets of S, then the relation

$\mathrm{R} = \{(\mathrm{A}, \mathrm{B}) : \mathrm{A} \cap \mathrm{B} \neq \phi; \mathrm{A}, \mathrm{B} \in \mathrm{M}\}$ is :

(A) symmetric only (B) reflexive only

(C) symmetric and reflexive only (D) symmetric and transitive only

Open solution

Q37. JEE Main 2023 (Online) 11th April Evening Shift

Let $\mathrm{A} = \{1, 3, 4, 6, 9\}$ and $\mathrm{B} = \{2, 4, 5, 8, 10\}$. Let R be a relation defined on $\mathrm{A} \times \mathrm{B}$ such that $\mathrm{R} = \{((a_1, b_1), (a_2, b_2)) : a_1 \leq b_2$ and $b_1 \leq a_2\}$. Then the number of elements in the set R is :

(A) 180 (B) 26 (C) 52 (D) 160

Open solution

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Q38. JEE Main 2023 (Online) 11th April Morning Shift

An organization awarded 48 medals in event 'A', 25 in event 'B' and 18 in event 'C'. If these medals went to total 60 men and only five men got medals in all the three events, then, how many received medals in exactly two of three events?

Open solution

(A) 10 (B) 15 (C) 21 (D) 9

Q39. JEE Main 2023 (Online) 10th April Evening Shift

Let $\mathrm{A}=\{2,3,4\}$ and $\mathrm{B}=\{8,9,12\}$. Then the number of elements in the relation $\mathrm{R}=\{((a_1,\ \mathrm{b}_1),(a_2,\ \mathrm{b}_2)) \in (A \times B, A \times B) : a_1$ divides b_2 and a_2 divides $\mathrm{b}_1\}$ is :

Open solution

(A) 18 (B) 24 (C) 36 (D) 12

Q40. JEE Main 2023 (Online) 8th April Evening Shift

Let $\mathrm{A}=\{1,2,3,4,5,6,7\}$. Then the relation $\mathrm{R}=\{(x, y) \in \mathrm{A} \times \mathrm{A} : x+y=7\}$ is :

Open solution

(A) reflexive but neither symmetric nor transitive

(B) transitive but neither symmetric nor reflexive

(C) symmetric but neither reflexive nor transitive

(D) an equivalence relation

Q41. JEE Main 2023 (Online) 1st February Evening Shift

Let $P(S)$ denote the power set of $S=\{1,2,3, \ldots, 10\}$. Define the relations R_1 and R_2 on $P(S)$ as $\mathrm{AR}_1 \mathrm{B}$ if $(\mathrm{A} \cap \mathrm{B}^{\mathrm{c}}) \cup (\mathrm{B} \cap \mathrm{A}^{\mathrm{c}})=\emptyset$ and $\mathrm{AR}_2 \mathrm{B}$ if $\mathrm{A} \cup \mathrm{B}^{\mathrm{c}}=\mathrm{B} \cup \mathrm{A}^{\mathrm{c}}, \forall \mathrm{A}, \mathrm{B} \in \mathrm{P}(\mathrm{S})$. Then :

Open solution

(A) only R_2 is an equivalence relation

(B) both R_1 and R_2 are not equivalence relations

(C) both R_1 and R_2 are equivalence relations

(D) only R_1 is an equivalence relation

Q42. JEE Main 2023 (Online) 1st February Morning Shift

Let R be a relation on $\mathbb{R}$, given by $R=\{(a, b) : 3a-3b+\sqrt{7}$ is an irrational number $\}$. Then R is

Open solution

(A) an equivalence relation

(B) reflexive and symmetric but not transitive

(C) reflexive and transitive but not symmetric

(D) reflexive but neither symmetric nor transitive

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Q43. JEE Main 2023 (Online) 31st January Evening Shift

Among the relations

$S = \{(a, b) : a, b \in \mathbb{R} - \{0\}, 2 + \frac{a}{b} > 0\}$

and $T = \{(a, b) : a, b \in \mathbb{R}, a^2 - b^2 \in \mathbb{Z}\}$,

(A) S is transitive but T is not
(B) both S and T are symmetric
(C) neither S nor T is transitive
(D) T is symmetric but S is not

Open solution

Q44. JEE Main 2023 (Online) 31st January Morning Shift

Let R be a relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)\ R\ (c, d)$ if and only if $ad(b - c) = bc(a - d)$. Then R is

(A) symmetric and transitive but not reflexive
(B) reflexive and symmetric but not transitive
(C) transitive but neither reflexive nor symmetric
(D) symmetric but neither reflexive nor transitive

Open solution

Q45. JEE Main 2023 (Online) 30th January Morning Shift

The minimum number of elements that must be added to the relation $R = \{(a, b), (b, c)\}$ on the set $\{a, b, c\}$ so that it becomes symmetric and transitive is :

(A) 7 (B) 3 (C) 4 (D) 5

Open solution

Q46. JEE Main 2023 (Online) 29th January Evening Shift

Let R be a relation defined on $\mathbb{N}$ as aRb if $2a + 3b$ is a multiple of $5, a, b \in \mathbb{N}$. Then R is

(A) an equivalence relation
(B) non reflexive
(C) symmetric but not transitive
(D) transitive but not symmetric

Open solution

Q47. JEE Main 2023 (Online) 24th January Morning Shift

The relation $R = \{(a, b) : \gcd(a, b) = 1, 2a \neq b, a, b \in \mathbb{Z}\}$ is :

(A) reflexive but not symmetric
(B) transitive but not reflexive
(C) symmetric but not transitive
(D) neither symmetric nor transitive

Open solution

Q48. JEE Main 2022 (Online) 29th July Morning Shift

Let R be a relation from the set $\{1, 2, 3, \ldots, 60\}$ to itself such that $R = \{(a, b) : b = pq$, where $p, q \geq 3$ are prime numbers$\}$. Then, the number of elements in R is :

(A) 600 (B) 660 (C) 540 (D) 720

Open solution

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Q49. JEE Main 2022 (Online) 28th July Morning Shift

For $\alpha \in \mathbf{N}$, consider a relation R on $\mathbf{N}$ given by $\mathrm{R} = \{(x, y) : 3x + \alpha y$ is a multiple of $7\}$. The relation R is an equivalence relation if and only if :

Open solution

(A) $\alpha = 14$

(B) α is a multiple of 4

(C) 4 is the remainder when α is divided by 10

(D) 4 is the remainder when α is divided by 7

Q50. JEE Main 2022 (Online) 27th July Morning Shift

Let R_1 and R_2 be two relations defined on $\mathbb{R}$ by

$a\,R_1\,b \Leftrightarrow ab \geq 0$ and $a\,R_2\,b \Leftrightarrow a \geq b$

Then,

Open solution

(A) R_1 is an equivalence relation but not R_2

(B) R_2 is an equivalence relation but not R_1

(C) both R_1 and R_2 are equivalence relations

(D) neither R_1 nor R_2 is an equivalence relation

Q51. JEE Main 2022 (Online) 29th June Evening Shift

The probability that a relation R from {x, y} to {x, y} is both symmetric and transitive, is equal to :

Open solution

(A) $\frac{5}{16}$ (B) $\frac{9}{16}$ (C) $\frac{11}{16}$ (D) $\frac{13}{16}$

Q52. JEE Main 2022 (Online) 29th June Morning Shift

Let a set $A = A_1 \cup A_2 \cup \ldots\ldots \cup A_k$, where $A_i \cap A_j = \phi$ for $i \neq j$, $1 \leq i, j \leq k$. Define the relation R from A to A by $R = \{(x, y) : y \in A_i$ if and only if $x \in A_i$, $1 \leq i \leq k\}$. Then, R is :

Open solution

(A) reflexive, symmetric but not transitive.

(B) reflexive, transitive but not symmetric.

(C) reflexive but not symmetric and transitive.

(D) an equivalence relation.

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Q53. JEE Main 2022 (Online) 28th June Evening Shift

Let $R_1 = \{(a, b) \in N \times N : |a - b| \leq 13\}$ and

$R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$. Then on N :

(A) Both R_1 and R_2 are equivalence relations

(B) Neither R_1 nor R_2 is an equivalence relation

(C) R_1 is an equivalence relation but R_2 is not

(D) R_2 is an equivalence relation but R_1 is not

Open solution

Q54. JEE Main 2021 (Online) 31st August Morning Shift

Which of the following is not correct for relation R on the set of real numbers ?

(A) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.

(B) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric and transitive.

(C) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.

(D) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric.

Open solution

Q55. JEE Main 2021 (Online) 27th August Evening Shift

Let Z be the set of all integers,

$$A = \{(x, y) \in Z \times Z : (x - 2)^2 + y^2 \leq 4\}$$

$$B = \{(x, y) \in Z \times Z : x^2 + y^2 \leq 4\}$$

$$C = \{(x, y) \in Z \times Z : (x - 2)^2 + (y - 2)^2 \leq 4\}$$

If the total number of relation from $A \cap B$ to $A \cap C$ is 2^p, then the value of p is :

(A) 16 (B) 25 (C) 49 (D) 9

Open solution

Q56. JEE Main 2021 (Online) 26th August Morning Shift

Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If K% of them are suffering from both ailments, then K can not belong to the set :

(A) {80, 83, 86, 89} (B) {84, 86, 88, 90} (C) {79, 81, 83, 85} (D) {84, 87, 90, 93}

Open solution

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Q57. JEE Main 2021 (Online) 27th July Evening Shift

Let N be the set of natural numbers and a relation R on N be defined by $R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is :

Open solution

(A) symmetric but neither reflexive nor transitive

(B) reflexive but neither symmetric nor transitive

(C) reflexive and symmetric, but not transitive

(D) an equivalence relation

Q58. JEE Main 2021 (Online) 18th March Evening Shift

Define a relation R over a class of n $\times$ n real matrices A and B as

"ARB iff there exists a non-singular matrix P such that PAP^{-1} = B".

Then which of the following is true?

Open solution

(A) R is reflexive, transitive but not symmetric

(B) R is symmetric, transitive but not reflexive.

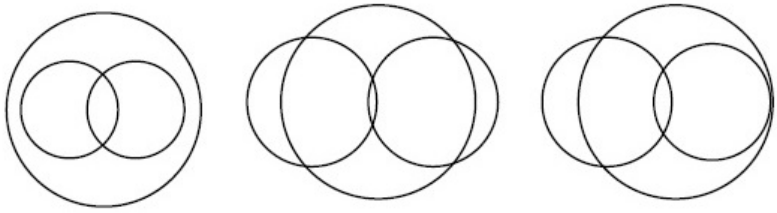
(C) R is reflexive, symmetric but not transitive

(D) R is an equivalence relation

Q59. JEE Main 2021 (Online) 17th March Morning Shift

In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement?

Open solution



(A) Q and R (B) None of these (C) P and R (D) P and Q

Q60. JEE Main 2021 (Online) 16th March Evening Shift

Let A = {2, 3, 4, 5, ..., 30} and '$\simeq$' be an equivalence relation on A $\times$ A, defined by (a, b) $\simeq$ (c, d), if and only if ad = bc. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair (4, 3) is equal to :

Open solution

(A) 5 (B) 6 (C) 8 (D) 7

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Q61. JEE Main 2021 (Online) 26th February Morning Shift

Let R = {(P, Q) | P and Q are at the same distance from the origin} be a relation, then the equivalence class of (1, −1) is the set :

Open solution

(A) $S = \{(x, y) | x^2 + y^2 = \sqrt{2}\}$

(B) $S = \{(x, y) | x^2 + y^2 = 2\}$

(C) $S = \{(x, y) | x^2 + y^2 = 1\}$

(D) $S = \{(x, y) | x^2 + y^2 = 4\}$

Q62. JEE Main 2020 (Online) 5th September Morning Slot

A survey shows that 73% of the persons working in an office like coffee, whereas 65% like tea. If x denotes the percentage of them, who like both coffee and tea, then x cannot be :

Open solution

(A) 63 (B) 36 (C) 54 (D) 38

Q63. JEE Main 2020 (Online) 4th September Evening Slot

Let $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^{n} Y_i = T$ where each X_i contains 10 elements and each Y_i contains 5 elements. If each element of the set T is an element of exactly 20 of sets X_i's and exactly 6 of sets Y_i's, then n is equal to :

Open solution

(A) 30 (B) 50 (C) 15 (D) 45

Q64. JEE Main 2020 (Online) 4th September Morning Slot

A survey shows that 63% of the people in a city read newspaper A whereas 76% read newspaper B. If x% of the people read both the newspapers, then a possible value of x can be:

Open solution

(A) 37 (B) 65 (C) 29 (D) 55

Q65. JEE Main 2020 (Online) 3rd September Evening Slot

Let R_1 and R_2 be two relation defined as follows :

$R_1 = \{(a, b) \in R^2 : a^2 + b^2 \in Q\}$ and

$R_2 = \{(a, b) \in R^2 : a^2 + b^2 \notin Q\}$,

where Q is the set of all rational numbers. Then :

Open solution

(A) Neither R_1 nor R_2 is transitive.

(B) R_2 is transitive but R_1 is not transitive.

(C) R_1 and R_2 are both transitive.

(D) R_1 is transitive but R_2 is not transitive.

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Q66. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

Consider the two sets :
$A = \{m \in R : \text{both the roots of } x^2 - (m + 1)x + m + 4 = 0 \text{ are real}\}$
and $B = [-3, 5)$.
Which of the following is not true?

(A) $A \cap B = \{-3\}$

(B) $B - A = (-3, 5)$

(C) $A \cup B = R$

(D) $A - B = (-\infty, -3) \cup (5, \infty)$

Q67. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

If $R = \{(x, y) : x, y \in Z, x^2 + 3y^2 \leq 8\}$ is a relation on the set of integers Z, then the domain of R^{-1} is :

(A) $\{0, 1\}$

(B) $\{-2, -1, 1, 2\}$

(C) $\{-1, 0, 1\}$

(D) $\{-2, -1, 0, 1, 2\}$

Q68. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

If $A = \{x \in R : |x| < 2\}$ and $B = \{x \in R : |x - 2| \geq 3\}$; then :

(A) $A - B = [-1, 2)$

(B) $A \cup B = R - (2, 5)$

(C) $A \cap B = (-2, -1)$

(D) $B - A = R - (-2, 5)$

Q69. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

Let A, B and C be sets such that $\phi \neq A \cap B \subseteq C$. Then which of the following statements is not true ?

(A) If $(A - B) \subseteq C$, then $A \subseteq C$

(B) $B \cap C \neq \phi$

(C) $(C \cup A) \cap (C \cup B) = C$

(D) If $(A - C) \subseteq B$, then $A \subseteq B$

Q70. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

Two newspapers A and B are published in a city. It is known that 25% of the city populations reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisement is :-

(A) 13.5

(B) 13

(C) 12.8

(D) 13.9

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Q71. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

Let Z be the set of integers.

If $A = \{x \in Z : 2^{(x+2)}(x^2 - 5x + 6)} = 1\}$ and

$B = \{x \in Z : -3 < 2x - 1 < 9\}$,

then the number of subsets of the set $A \times B$, is

(A) 2^{12} (B) 2^{18} (C) 2^{10} (D) 2^{15}

Q72. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

Let S = {1, 2, 3, ... , 100}. The number of non-empty subsets A of S such that the product of elements in A is even is :

(A) $2^{50} - 1$ (B) $2^{50}(2^{50} - 1)$ (C) $2^{100} - 1$ (D) $2^{50} + 1$

Q73. JEE Main 2019 (Online) 10th January Morning Slot

Open solution

In a class of 140 students numbered 1 to 140, all even numbered students opted Mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is

(A) 42 (B) 102 (C) 1 (D) 38

Q74. JEE Main 2018 (Online) 16th April Morning Slot

Open solution

Let **N** denote the set of all natural numbers. Define two binary relations on **N** as $R_1 = \{(x, y) \in \mathbf{N} \times \mathbf{N} : 2x + y = 10\}$ and $R_2 = \{(x, y) \in \mathbf{N} \times \mathbf{N} : x + 2y = 10\}$. Then :

(A) Range of R_1 is {2, 4, 8}. (B) Range of R_2 is {1, 2, 3, 4}.

(C) Both R_1 and R_2 are symmetric relations. (D) Both R_1 and R_2 are transitive relations.

Q75. JEE Main 2018 (Offline)

Open solution

Two sets A and B are as under :

$A = \{(a, b) \in R \times R : |a - 5| < 1 \text{ and } |b - 5| < 1\}$;

$B = \{(a, b) \in R \times R : 4(a - 6)^2 + 9(b - 5)^2 \leq 36\}$;

Then

(A) neither $A \subset B$ nor $B \subset A$ (B) $B \subset A$

(C) $A \subset B$ (D) $A \cap B = \phi$ (an empty set)

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Q76. JEE Main 2018 (Online) 15th April Morning Slot

Consider the following two binary relations on the set A = {a, b, c} :
R_1 = {(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)} and
R_2 = {(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)}.
Then :

(A) both R_1 and R_2 are not symmetric.

(B) R_1 is not symmetric but it is transitive.

(C) R_2 is symmetric but it is not transitive.

(D) both R_1 and R_2 are transitive.

Open solution

Q77. JEE Main 2015 (Online) 10th April Morning Slot

In a certain town, 25% of the families own a phone and 15% own a car ; 65% families own neither a phone nor a car and 2,000 families own both a car and a phone. Consider the following three statements :

(a) 5% families own both a car and a phone

(b) 35% families own either a car or a phone.

(c) 40,000 families live in the town.

Then,

(A) Only (a) and (b) are correct.

(B) Only (a) and (c) are correct.

(C) Only (b) and (c) are correct.

(D) All (a), (b) and (c) are correct.

Open solution

Q78. JEE Main 2015 (Offline)

Let A and B be two sets containing four and two elements respectively. Then, the number of subsets of the set A $\times$ B , each having at least three elements are

(A) 219 **(B)** 256 **(C)** 275 **(D)** 510

Open solution

Q79. JEE Main 2014 (Online) 12th April Morning Slot

A relation on the set A = {x : $|x|$ < 3, x$\in$Z}, where Z is the set of integers is defined by R = {(x, y) : y = $|x|$, x $\neq -1$}. Then the number of elements in the power set of R is :

(A) 32 **(B)** 16 **(C)** 8 **(D)** 64

Open solution

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Q80. JEE Main 2014 (Online) 9th April Morning Slot

Let P be the relation defined on the set of all real numbers such that
$P = \{(a, b) : \sec^2 a - \tan^2 b = 1\}$. Then P is :

(A) reflexive and symmetric but not transitive.

(B) reflexive and transitive but not symmetric.

(C) symmetric and transitive but not reflexive.

(D) an equivalence relation.

Open solution

Q81. JEE Main 2014 (Offline)

If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \{9(n-1) : n \in N\}$, where N is the set of natural numbers, then $X \cup Y$ is equal to:

(A) X (B) Y (C) N (D) $Y - X$

Open solution

Q82. JEE Main 2013 (Online) 23th April Morning Slot

Let R = {(x, y) : x, y ∈ N and $x^2 - 4xy + 3y^2 = 0$}, where N is the set of all natural numbers. Then the relation R is:

(A) reflexive but neither symmetric nor transitive

(B) symmetric and transitive

(C) reflexive and symmetric

(D) reflexive and transitive

Open solution

Q83. JEE Main 2013 (Online) 22th April Morning Slot

Let R = {(3, 3) (5, 5), (9, 9), (12, 12), (5, 12), (3, 9), (3, 12), (3, 5)} be a relation on the set A = {3, 5, 9, 12}. Then, R is:

(A) reflexive, symmetric but not transitive

(B) symmetric, transitive but not reflexive

(C) an equivalence relation

(D) reflexive, transitive but not symmetric

Open solution

Q84. AIEEE 2012 (Online) 12th May

If $A = \{x \in z^+ : x < 10$ and x is a multiple of 3 or $4\}$, where z^+ is the set of positive integers, then the total number of symmetric relations on A is

(A) 2^5 (B) 2^{15} (C) 2^{10} (D) 2^{20}

Open solution

Q85. AIEEE 2012

Let X = {1, 2, 3, 4, 5}. The number of different ordered pairs (Y, Z) that can be formed such that $Y \subseteq X$, $Z \subseteq X$ and $Y \cap Z$ is empty, is :

(A) 3^5 (B) 2^5 (C) 5^3 (D) 5^2

Open solution

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Q86. AIEEE 2011

Open solution

Let R be the set of real numbers.

Statement I : $A = \{(x, y) \in R \times R : y - x$ is an integer $\}$ is an equivalence relation on R.

Statement II : $B = \{(x, y) \in R \times R : x = \alpha y$ for some rational number $\alpha\}$ is an equivalence relation on R.

(A) Statement I is true, Statement II is true; Statement II is not a correct explanation for Statement I.

(B) Statement I is true, Statement II is false.

(C) Statement I is false, Statement II is true.

(D) Statement I is true, Statement II is true; Statement II is a correct explanation for Statement I.

Q87. AIEEE 2010

Open solution

Consider the following relations

$R = \{(x, y) \mid x, y$ are real numbers and $x = wy$ for some rational number $w\}$;

$S = \left\{\left(\frac{m}{n}, \frac{p}{q}\right) \mid m, n, p$ and q are integers such that $n, q \neq 0$ and $qm = pn\}$. Then

(A) R is an equivalence relation but S is not an equivalence relation

(B) Neither R nor S is an equivalence relation

(C) S is an equivalence relation but R is not an equivalence relation

(D) R and S both are equivalence relations

Q88. AIEEE 2009

Open solution

If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then :

(A) $A = C$ (B) $B = C$ (C) $A \cap B = \phi$ (D) $A = B$

Q89. AIEEE 2008

Open solution

Let R be the real line. Consider the following subsets of the plane $R \times R$:
$S = \{(x, y) : y = x + 1 \; and \; 0 < x < 2\}$
$T = \{(x, y) : x - y \; is \; an \; integer\}$,
Which one of the following is true ?

(A) Neither S nor T is an equivalence relation on R

(B) Both S and T are equivalence relation on R

(C) S is an equivalence relation on R but T is not

(D) T is an equivalence relation on R but S is not

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Q90. AIEEE 2006

Let W denote the words in the English dictionary. Define the relation R by

$R = \{(x, y) \in W \times W \mid$ the words x and y have at least one letter in common$\}$. Then, R is

(A) reflexive, symmetric and not transitive
(B) reflexive, symmetric and transitive
(C) reflexive, not symmetric and transitive
(D) not reflexive, symmetric and transitive

Open solution

Q91. AIEEE 2005

Let $R = \{(3,3), (6,6), (9,9), (12,12), (6,12), (3,9), (3,12), (3,6)\}$ be a relation on the set $A = \{3, 6, 9, 12\}$. The relation is :

(A) reflexive and symmetric only
(B) an equivalence relation
(C) reflexive only
(D) reflexive and transitive only

Open solution

Q92. AIEEE 2004

Let $R = \{(1,3), (4,2), (2,4), (2,3), (3,1)\}$ be a relation on the set $A = \{1,2,3,4\}$. The relation R is :

(A) a function
(B) transitive
(C) not symmetric
(D) reflexive

Open solution

Numerical 34 questions

Q93. JEE Main 2026 (Online) 6th April Evening Shift

Let $\mathrm{R} = \{(x, y) \in \mathbf{N} \times \mathbf{N} : \log_e(x + y) \leq 2\}$. Then the minimum number of elements, required to be added in R to make it a transitive relation, is ____ .

Open solution

Q94. JEE Main 2026 (Online) 5th April Evening Shift

Let $\mathrm{A} = \{1, 4, 7\}$ and $\mathrm{B} = \{2, 3, 8\}$. Then the number of elements, in the relation $R = \{((a_1, b_1), (a_2, b_2)) \in ((A \times B) \times (A \times B)) : a_1 + b_2$ divides $a_2 + b_1\}$ is ____ .

Open solution

Q95. JEE Main 2026 (Online) 2nd April Evening Shift

Let $A = \{2, 3, 4, 5, 6\}$. Let R be a relation on the set $A \times A$ given by $(x, y)R(z, w)$ if and only if x divides z and $y \leq w$. Then the number of elements in R is _____.

Open solution

Q96. JEE Main 2026 (Online) 22nd January Evening Shift

Let S be the set of the first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \geqslant 2$ and the product of all elements of B is even $\}$ is ____ .

Open solution

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Q97. JEE Main 2025 (Online) 7th April Morning Shift

The number of relations on the set $A = \{1, 2, 3\}$, containing at most 6 elements including $(1, 2)$, which are reflexive and transitive but not symmetric, is _____.

Open solution

Q98. JEE Main 2025 (Online) 7th April Morning Shift

For $n \geq 2$, let S_n denote the set of all subsets of $\{1, 2, \ldots, n\}$ with no two consecutive numbers. For example $\{1, 3, 5\} \in S_6$, but $\{1, 2, 4\} \notin S_6$. Then $n(S_5)$ is equal to _____

Open solution

Q99. JEE Main 2025 (Online) 24th January Morning Shift

Let $S = \{p_1, p_2 \ldots, p_{10}\}$ be the set of first ten prime numbers. Let $A = S \cup P$, where P is the set of all possible products of distinct elements of S. Then the number of all ordered pairs $(x, y), x \in S$, $y \in A$, such that x divides y, is _____ .

Open solution

Q100. JEE Main 2025 (Online) 22nd January Evening Shift

Let $A = \{1, 2, 3\}$. The number of relations on A, containing $(1, 2)$ and $(2, 3)$, which are reflexive and transitive but not symmetric, is _____.

Open solution

Q101. JEE Main 2024 (Online) 9th April Morning Shift

Let $A = \{2, 3, 6, 7\}$ and $B = \{4, 5, 6, 8\}$. Let R be a relation defined on $A \times B$ by $(a_1, b_1)R(a_2, b_2)$ if and only if $a_1 + a_2 = b_1 + b_2$. Then the number of elements in R is _____.

Open solution

Q102. JEE Main 2024 (Online) 4th April Morning Shift

In a survey of 220 students of a higher secondary school, it was found that at least 125 and at most 130 students studied Mathematics; at least 85 and at most 95 studied Physics; at least 75 and at most 90 studied Chemistry; 30 studied both Physics and Chemistry; 50 studied both Chemistry and Mathematics; 40 studied both Mathematics and Physics and 10 studied none of these subjects. Let m and n respectively be the least and the most number of students who studied all the three subjects. Then $\mathrm{m} + \mathrm{n}$ is equal to _____.

Open solution

Q103. JEE Main 2024 (Online) 1st February Morning Shift

Let $A = \{1, 2, 3, \ldots, 20\}$. Let R_1 and R_2 be two relations on A such that

$R_1 = \{(a, b) : b \text{ is divisible by } a\}$

$R_2 = \{(a, b) : a \text{ is an integral multiple of } b\}$.

Then, number of elements in $R_1 - R_2$ is equal to _____.

Open solution

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Q104. JEE Main 2024 (Online) 31st January Evening Shift

Let $A = \{1, 2, 3, \ldots\ldots\ldots\ldots, 100\}$. Let R be a relation on A defined by $(x, y) \in R$ if and only if $2x = 3y$. Let R_1 be a symmetric relation on A such that $R \subset R_1$ and the number of elements in R_1 is n. Then, the minimum value of n is _____.

Open solution

Q105. JEE Main 2024 (Online) 31st January Morning Shift

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (1, 4)\}$ be a relation on A. Let S be the equivalence relation on A such that $R \subset S$ and the number of elements in S is n. Then, the minimum value of n is _____.

Open solution

Q106. JEE Main 2024 (Online) 30th January Evening Shift

The number of symmetric relations defined on the set $\{1, 2, 3, 4\}$ which are not reflexive is _____.

Open solution

Q107. JEE Main 2023 (Online) 15th April Morning Shift

The number of elements in the set

$\{n \in \mathbb{N} : 10 \leq n \leq 100$ and $3^n - 3$ is a multiple of $7\}$ is _____.

Open solution

Q108. JEE Main 2023 (Online) 15th April Morning Shift

Let $A = \{1, 2, 3, 4\}$ and R be a relation on the set $A \times A$ defined by

$R = \{((a, b), (c, d)) : 2a + 3b = 4c + 5d\}$. Then the number of elements in R is _____.

Open solution

Q109. JEE Main 2023 (Online) 13th April Evening Shift

Let $\mathrm{A} = \{-4, -3, -2, 0, 1, 3, 4\}$ and $\mathrm{R} = \{(a, b) \in \mathrm{A} \times \mathrm{A} : b = |a|$ or $b^2 = a + 1\}$ be a relation on A. Then the minimum number of elements, that must be added to the relation R so that it becomes reflexive and symmetric, is _____

Open solution

Q110. JEE Main 2023 (Online) 12th April Morning Shift

The number of relations, on the set $\{1, 2, 3\}$ containing $(1, 2)$ and $(2, 3)$, which are reflexive and transitive but not symmetric, is _____.

Open solution

Q111. JEE Main 2023 (Online) 10th April Morning Shift

The number of elements in the set $\{n \in Z : |n^2 - 10n + 19| < 6\}$ is _____.

Open solution

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Q112. JEE Main 2023 (Online) 8th April Morning Shift

Let $A = \{0, 3, 4, 6, 7, 8, 9, 10\}$ and R be the relation defined on A such that $R = \{(x, y) \in A \times A : x - y$ is odd positive integer or $x - y = 2\}$. The minimum number of elements that must be added to the relation R, so that it is a symmetric relation, is equal to _____.

Open solution

Q113. JEE Main 2023 (Online) 6th April Morning Shift

Let $\mathrm{A} = \{1, 2, 3, 4, \ldots, 10\}$ and $\mathrm{B} = \{0, 1, 2, 3, 4\}$. The number of elements in the relation $R = \{(a, b) \in A \times A : 2(a - b)^2 + 3(a - b) \in B\}$ is _____.

Open solution

Q114. JEE Main 2023 (Online) 25th January Morning Shift

Let S = {1, 2, 3, 5, 7, 10, 11}. The number of non-empty subsets of S that have the sum of all elements a multiple of 3, is _____.

Open solution

Q115. JEE Main 2023 (Online) 24th January Evening Shift

The minimum number of elements that must be added to the relation R = {(a, b), (b, c), (b, d)} on the set {a, b, c, d} so that it is an equivalence relation, is _____.

Open solution

Q116. JEE Main 2022 (Online) 29th July Morning Shift

Let $S = \{4, 6, 9\}$ and $T = \{9, 10, 11, \ldots, 1000\}$. If $A = \{a_1 + a_2 + \ldots + a_k : k \in \mathbf{N}, a_1, a_2, a_3, \ldots, a_k \in S\}$, then the sum of all the elements in the set $T - A$ is equal to _____.

Open solution

Q117. JEE Main 2022 (Online) 26th July Evening Shift

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{3, 6, 7, 9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \phi\}$ is _____.

Open solution

Q118. JEE Main 2022 (Online) 25th July Evening Shift

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. Define $B = \{T \subseteq A$: either $1 \notin T$ or $2 \in T\}$ and $C = \{T \subseteq A$: the sum of all the elements of T is a prime number $\}$. Then the number of elements in the set $B \cup C$ is _____.

Open solution

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Q119. JEE Main 2022 (Online) 28th June Morning Shift

Let R_1 and R_2 be relations on the set {1, 2, ..., 50} such that

$R_1 = \{(p, p^n) : p \text{ is a prime and } n \geq 0 \text{ is an integer}\}$ and

$R_2 = \{(p, p^n) : p \text{ is a prime and } n = 0 \text{ or } 1\}$.

Then, the number of elements in $R_1 - R_2$ is _____.

Open solution

Q120. JEE Main 2022 (Online) 26th June Morning Shift

Let $A = \{n \in N : \text{H.C.F. } (n, 45) = 1\}$ and

Let $B = \{2k : k \in \{1, 2, ..., 100\}\}$. Then the sum of all the elements of $A \cap B$ is _____.

Open solution

Q121. JEE Main 2022 (Online) 24th June Evening Shift

The sum of all the elements of the set $\{\alpha \in \{1, 2, ..., 100\} : HCF(\alpha, 24) = 1\}$ is _____.

Open solution

Q122. JEE Main 2021 (Online) 27th August Morning Shift

If $A = \{x \in R : |x - 2| > 1\}$,
$B = \{x \in R : \sqrt{x^2 - 3} > 1\}$,
$C = \{x \in R : |x - 4| \geq 2\}$ and Z is the set of all integers, then the number of subsets of the set $(A \cap B \cap C)^c \cap Z$ is _____.

Open solution

Q123. JEE Main 2021 (Online) 27th July Evening Shift

Let $A = \{n \in N \mid n^2 \leq n + 10{,}000\}$, $B = \{3k + 1 \mid k \in N\}$ and $C = \{2k \mid k \in N\}$, then the sum of all the elements of the set $A \cap (B - C)$ is equal to _____.

Open solution

Q124. JEE Main 2021 (Online) 24th February Morning Shift

Let $A = \{n \in N: n \text{ is a 3-digit number}\}$

$B = \{9k + 2: k \in N\}$

and $C = \{9k + l: k \in N\}$ for some $l(0 < l < 9)$

If the sum of all the elements of the set $A \cap (B \cup C)$ is 274×400, then l is equal to _____.

Open solution

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Q125. JEE Main 2020 (Online) 6th September Morning Slot

Set A has m elements and set B has n elements. If the total number of subsets of A is 112 more than the total number of subsets of B, then the value of m.n is _____.

Open solution

Q126. JEE Main 2020 (Online) 7th January Evening Slot

Let X = {n $\in$ N : 1 $\leq$ n $\leq$ 50}. If
A = {n $\in$ X: n is a multiple of 2} and
B = {n $\in$ X: n is a multiple of 7}, then the number of elements in the smallest subset of X containing both A and B is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **C**	3. **D**	4. **C**	5. **D**
6. **D**	7. **B**	8. **B**	9. **B**	10. **C**
11. **B**	12. **A**	13. **C**	14. **B**	15. **A**
16. **D**	17. **B**	18. **D**	19. **D**	20. **C**
21. **B**	22. **B**	23. **D**	24. **C**	25. **B**
26. **C**	27. **C**	28. **C**	29. **A**	30. **D**
31. **D**	32. **C**	33. **B**	34. **B**	35. **B**
36. **A**	37. **D**	38. **C**	39. **C**	40. **C**
41. **C**	42. **D**	43. **D**	44. **D**	45. **A**
46. **A**	47. **D**	48. **B**	49. **D**	50. **D**
51. **A**	52. **D**	53. **B**	54. **B**	55. **B**
56. **C**	57. **B**	58. **D**	59. **B**	60. **D**
61. **B**	62. **B**	63. **A**	64. **D**	65. **A**
66. **D**	67. **C**	68. **D**	69. **D**	70. **D**
71. **D**	72. **B**	73. **D**	74. **B**	75. **C**
76. **C**	77. **D**	78. **A**	79. **B**	80. **D**
81. **B**	82. **A**	83. **D**	84. **B**	85. **A**
86. **B**	87. **C**	88. **B**	89. **D**	90. **A**
91. **D**	92. **C**			

Numerical

93. **15**	94. **18**	95. **120**	96. **1979**	97. **6**
98. **13**	99. **5120**	100. **3**	101. **25**	102. **45**
103. **46**	104. **66**	105. **16**	106. **960**	107. **15**
108. **6**	109. **7**	110. **3**	111. **6**	112. **19**
113. **18**	114. **43**	115. **13**	116. **11**	117. **112**
118. **107**	119. **8**	120. **5264**	121. **1633**	122. **256**
123. **832**	124. **5**	125. **28**	126. **29**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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